IGCSE 0607 Practice Questions — E2.2 — Combined Questions & Markscheme

Algebraic manipulation

Total Questions: 12 — **Total Marks:** 69 Marks

Table of Contents & Quick Navigation

- **E2.2** Algebraic manipulation
  → **Paper 2 (Non-Calculator)** *(9 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

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1 **(a)** Expand and simplify.

$3(2x - 5) - 2(x + 4)$

..... **[2]**

(b) Factorise completely.

$8x^2y - 12xy^2$

..... **[2]**

2 **(a)** Expand and simplify

$$3(2x - 5) - 4(x - 2)$$

..... **[2]**

(b) Factorise fully

$$12x^2y - 18xy^2$$

..... **[2]**

3 **(a)** Expand and simplify.

$3(2x - 5) - 4(x - 3)$

..... **[2]**

(b) Factorise fully.

$14x^2y + 21xy^2$

..... **[2]**

4 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$ac - 3bc + 2ad - 6bd$

..... **[2]**

(c) Factorise.

$16x^2 - 25y^2$

..... **[1]**

5 **(a)** Factorise.

$x^2 - 8x + 15$

..... **[2]**

(b) Factorise.

$2x^2 - 5x - 12$

..... **[2]**

(c) Factorise.

$ax - 3a + 2bx - 6b$

..... **[2]**

6 **(a)** Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

(b) Factorise completely.

$3x^3 + 10x^2 - 8x$

..... **[3]**

7 **(a)** Factorise

$$6ax - 2ay - 3bx + by$$

..... **[2]**

(b) Factorise completely.

(i)

$$50x^2 - 8y^2$$

..... **[3]**

(ii)

$$2x^3 + 5x^2 - 12x$$

..... **[3]**

8 **(a)** Expand and simplify

$$(x-3)(2x+1)(x+4)$$

..... **[3]**

(b) Factorise

$$2x^2-7x-15$$

..... **[2]**

9 **(a)** Expand and simplify.

$(x-2)(x+3)(2x-1)$

..... **[3]**

(b) Factorise.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$2x^3 - 18x$

..... **[3]**

10 **(a)** Expand and simplify.

$4(2x - 3y) - 3(x - 5y)$

..... **[2]**

(b) Factorise completely.

$14a^2b - 21ab^2$

..... **[2]**

11 **(a)** Expand and simplify.

$(2x - 5)(3x + 1)$

..... **[2]**

(b) Factorise.

$x^2 - 7x - 18$

..... **[2]**

(c) Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

12 **(a)** Expand and simplify.

$(x-4)(2x+3)(x+1)$

..... **[3]**

(b) Factorise completely.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$x^3 + 3x^2 - 4x - 12$

..... **[3]**

1 **(a)** $6x - 15 - 2x - 8$ **[M1]**

$4x - 23$ **[A1]**

(b) $4xy(2x - 3y)$ or $4x(2xy - 3y^2)$ or $4y(2x^2 - 3xy)$ or $xy(8x - 12y)$ **[M1]**

$4xy(2x - 3y)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Correct expansion: $6x - 15 - 4x + 8$ **[M1]**

$2x - 7$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

Correct partial factorisation e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$ **[M1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** Expansion to give $6x - 15$ and $-4x + 12$ **[M1]**

$2x - 3$ **[A1]**

(b) $7xy(2x + 3y)$ **[B2]**

Partial factorisation, e.g. $7x(2xy + 3y^2)$ or $xy(14x + 21y)$ **[B1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $c(a - 3b) + 2d(a - 3b)$ or $a(c + 2d) - 3b(c + 2d)$ **[M1]**

$(a - 3b)(c + 2d)$ **[A1]**

(c) $(4x - 5y)(4x + 5y)$ **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $(x + a)(x + b)$ where $ab = 15$ or $a + b = -8$ **[M1]**

$(x - 3)(x - 5)$ **[A1]**

(b) $(2x + a)(x + b)$ where $ab = -12$ or $2b + a = -5$ **[M1]**

$(2x + 3)(x - 4)$ **[A1]**

(c) $a(x - 3) + 2b(x - 3)$ or $x(a + 2b) - 3(a + 2b)$ **[M1]**

$(a + 2b)(x - 3)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** Taking out common factor to reach $2(9x^2 - 25y^2)$ **[M1]**

Factorising difference of squares to reach $(3x-5y)(3x+5y)$ **[M1]**

$2(3x-5y)(3x+5y)$ **[A1]**

(b) Taking out common factor x to reach $x(3x^2+10x-8)$ **[M1]**

Factorising quadratic to reach $(3x-2)(x+4)$ **[M1]**

$x(3x-2)(x+4)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

7 (a) $(2a-b)(3x-y)$ **[B2]**

Correct partial factorisation e.g., $2a(3x-y)-b(3x-y)$ or $3x(2a-b)-y(2a-b)$ **[M1]**

(b) (i) Extracts common factor of 2: $2(25x^2-4y^2)$ **[M1]**

Factorises difference of two squares: $(5x-2y)(5x+2y)$ **[M1]**

$2(5x-2y)(5x+2y)$ **[A1]**

(ii) Extracts common factor of x: $x(2x^2+5x-12)$ **[M1]**

Factorises quadratic: $(2x-3)(x+4)$ **[M1]**

$x(2x-3)(x+4)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

8 (a) Correct expansion of any two brackets, e.g., $(2x^2-5x-3)$ or $(2x^2+9x+4)$ or (x^2+x-12) **[M1]**

Correct expansion of their quadratic with the third bracket, producing at least 5 terms, e.g., $2x^3+8x^2-5x^2-20x-3x-12$ **[M1]**

$2x^3+3x^2-23x-12$ **[A1]**

(b) $(2x+3)(x-5)$ **[B2]**

Accept $(2x+a)(x+b)$ where $ab=-15$ or $2b+a=-7$ **[M1]**

[Back to Question] *[Total Marks: 5]*

9 (a) $(x^2+x-6)(2x-1)$ or $(x-2)(2x^2+5x-3)$ or $(x+3)(2x^2-5x+2)$ **[M1]**

$2x^3-x^2+2x^2-x-12x+6$ (allow one error in expansion) **[M1]**

$2x^3+x^2-13x+6$ **[A1]**

(b) $(3x\pm a)(2x\pm b)$ where $ab=10$ or $6x^2-15x+4x-10$ **[M1]**

$(3x+2)(2x-5)$ **[A1]**

(c) $2x(x^2-9)$ **[M1]**

$(x-3)(x+3)$ seen **[M1]**

$2x(x-3)(x+3)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

10 **(a)** $8x - 12y - 3x + 15y$ **[M1]**

$5x + 3y$ **[A1]**

(b) $7ab$ extracted as a common factor **[M1]**

$7ab(2a - 3b)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** $6x^2 + 2x - 15x - 5$ (at least 3 correct terms) **[M1]**

$6x^2 - 13x - 5$ **[A1]**

(b) $(x + a)(x + b)$ where $ab = -18$ or $a + b = -7$ **[M1]**

$(x - 9)(x + 2)$ **[A1]**

(c) $2(9x^2 - 25y^2)$ **[M1]**

$(3x - 5y)(3x + 5y)$ seen **[M1]**

$2(3x - 5y)(3x + 5y)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

12 **(a)** Expands any two brackets correctly, e.g., $(2x^2 - 5x - 12)(x + 1)$ or $(x - 4)(2x^2 + 5x + 3)$ or $(x^2 - 3x - 4)(2x + 3)$ **[M1]**

Multiplies their resulting quadratic by the third bracket, allowing one error **[M1]**

$2x^3 - 3x^2 - 17x - 12$ **[A1]**

(b) $(3x + a)(2x + b)$ or $(6x + c)(x + d)$ where $ab = -10$ or $cd = -10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $x^2(x + 3) - 4(x + 3)$ or $x(x^2 - 4) + 3(x^2 - 4)$ **[M1]**

$(x^2 - 4)(x + 3)$ **[M1]**

$(x - 2)(x + 2)(x + 3)$ **[A1]**

[Back to Question] *[Total Marks: 8]*